Nepal Electoral Constituency GeoJSON

Scraping SVG, converting to GIS, and Georeferencing data with R

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1 Introduction

In this workflow, we tackle a common geospatial problem: extracting data from stylized news graphics (like election heatmaps) and converting them into GIS-ready formats.

Rather than just cleaning the rough edges of a scraped SVG, we will use a **reconstruction technique**. We will:

1. **Scrape** the electoral map to get rough constituency shapes.
2. **Georeferencing** the shapes to align them with real coordinates.
3. **Reconstruct** the precise boundaries by grouping official administrative "Wards" based on which constituency shape they fall into.

2 1. Setup and Libraries

We use `sf` for spatial operations and `rvest` for scraping. We also define paths for our inputs (official ward data) and outputs.

```
library(sf)       # Spatial data handling
library(xml2)     # Parsing XML/SVG
library(rvest)    # Web scraping
library(httr2)    # HTTP requests
```

```
library(stringr)  # String manipulation
library(dplyr)    # Data manipulation
library(here)     # Robust project-relative paths

# We build the path to the current post directory.
post_dir <- here("posts", "nepal-election-geojson")

# 2. Define paths to data folder
wards_geojson_path    <- here::here(post_dir, "data", "nepal-wards.gpkg")
national_geojson_path <- here::here(post_dir, "data", "nepal-national.gpkg")

# 3. Define output paths
output_svg_path                          <-    here::here(post_dir,    "output",
"nepal_electoral_map.svg")
rough_map_path <- here::here(post_dir, "output", "nepal_svg_rough.geojson")
output_geojson_path                 <-      here::here(post_dir,      "output",
"nepal_electoral_map.geojson")

# Ensure output directory exists
if (!dir.exists(dirname(output_svg_path))) dir.create(dirname(output_svg_path),
recursive = TRUE)
```

3 Scraping the SVG

We use `httr2` to fetch the webpage and `rvest` to extract the specific SVG element containing the map. We also inject some CSS to ensure the map renders correctly if viewed in a browser (e.g., setting the viewBox and handling overflow).

```
url <- "https://generalelection2079.ekantipur.com/heatmap"

message("Fetching data from: ", url)

tryCatch({
  # 1. Fetch Content
  req <- request(url) |>
            req_user_agent("Mozilla/5.0   (Windows   NT   10.0;   Win64;   x64)
AppleWebKit/537.36")

  resp <- req_perform(req)
  html_content <- resp_body_html(resp)

  # 2. Extract SVG (Select by pattern ID specific to this map)
       svg_node    <-    html_element(html_content,    xpath    =    "//svg[.//
```

```
pattern[@id='np__BG']]")

  if (!inherits(svg_node, "xml_missing")) {
    # Fix viewBox for consistency (1920x1080)
    xml_set_attr(svg_node, "viewBox", "0 0 1920 1080")

    # Add CSS for better visibility
    css_styles <- "
     path { fill: #cccccc; stroke: #fff; cursor: pointer; stroke-width: 1.75px; }
      path:hover { opacity: 0.9; }
      .national-park { fill: #55e5a5 !important; }
      svg { overflow: visible !important; }
    "
    xml_add_child(svg_node, "style", css_styles, .where = 0)

    # Save to output folder
    write_xml(svg_node, output_svg_path)
    message("Success! Raw SVG saved to: ", output_svg_path)

  } else {
    warning("SVG element not found on the page.")
  }
}, error = function(e) {
  message("Error fetching URL: ", e$message)
})
```

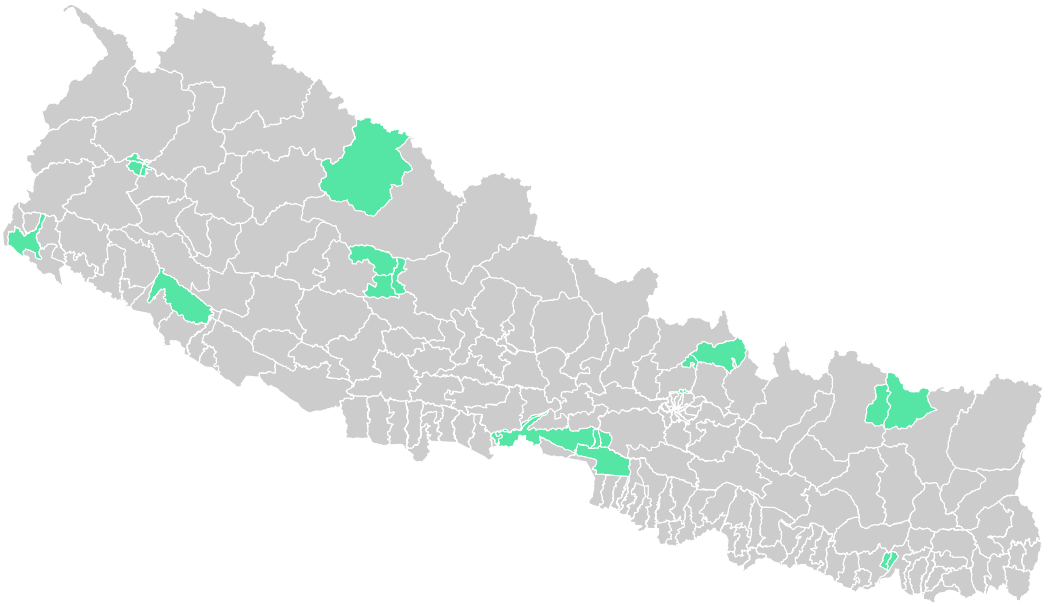


Figure 1: Scraped Electoral Map of Nepal in SVG format

4 Parsing SVG Paths to Geometry

SVG paths are stored as strings in the `d` attribute. We need a robust parser to convert these strings into numerical coordinates. This function handles messy spacing and implicit commands often found in web SVGs.

```
# Function to robustly handle SVG path strings (fixes minified negatives)
parse_d_robust <- function(d) {
  d_clean <- str_replace_all(d, "-", " -")             # Handle "100-100"
  d_clean <- str_replace_all(d_clean, "[a-zA-Z]", " ") # Remove commands
  nums <- as.numeric(str_split(d_clean, "[\\s,]+")[[1]])
  nums <- nums[!is.na(nums)]

  if(length(nums) < 4) return(NULL)

  mtx <- matrix(nums, ncol = 2, byrow = TRUE)
  # Close polygon if open
  if (!all(mtx[1,] == mtx[nrow(mtx),])) mtx <- rbind(mtx, mtx[1,])
  if (nrow(mtx) < 4) return(NULL)

  return(mtx)
}

# Extract paths from the scraped SVG content
```

```
path_nodes <- xml_find_all(svg_node, "//*[local-name()='path']")

polys_list <- list()
id_list <- character()

for (i in seq_along(path_nodes)) {
  d_attr <- xml_attr(path_nodes[i], "d")
  id_val <- xml_attr(path_nodes[i], "data-slug")

  # --- UI FILTER FIX ---
  # Strictly skip paths that lack a data-slug (buttons, background rects)
  if (is.na(id_val) || id_val == "") next

  coords <- parse_d_robust(d_attr)
  if (!is.null(coords)) {
    polys_list[[length(polys_list) + 1]] <- st_polygon(list(coords))
    id_list <- c(id_list, id_val)
  }
}

if (length(polys_list) == 0) stop("No valid map districts found after
filtering.")

# Create initial SF object (Raw SVG Coordinates)
nepal_sf <- st_sf(constituency_id = id_list, geometry = st_sfc(polys_list))
```

5 Georeferencing (Affine Transformation)

The extracted SVG uses screen coordinates (pixels). To make this map useful, we must align it with the real world (Longitude/Latitude).

We do this by calculating the bounding box of a reference map (`data/nepal-national.geojson`) and scaling our SVG to fit those dimensions.

```
# 1. Load Reference Map
if (!file.exists(national_geojson_path)) stop("Reference map not found.")

nepal_ref <- st_read(national_geojson_path, quiet = TRUE)

# 2. Project Reference to Web Mercator (EPSG:3857)
# This gives us the correct target aspect ratio for a web map
target_bbox <- st_bbox(st_transform(nepal_ref, 3857))

# 3. Affine Transformation
```

```r
# Flip Y-axis (SVG top-left origin -> Map bottom-left origin)
flip_matrix <- matrix(c(1, 0, 0, -1), ncol = 2)
nepal_sf$geometry <- nepal_sf$geometry * flip_matrix

# Calculate Scale Factors
src_bbox <- st_bbox(nepal_sf)
src_w <- src_bbox$xmax - src_bbox$xmin
src_h <- src_bbox$ymax - src_bbox$ymin

tgt_w <- target_bbox$xmax - target_bbox$xmin
tgt_h <- target_bbox$ymax - target_bbox$ymin

# Shift SVG to (0,0)
nepal_sf$geometry <- nepal_sf$geometry - c(src_bbox$xmin, src_bbox$ymin)

# Scale to match Reference size
scale_matrix <- matrix(c(tgt_w / src_w, 0, 0, tgt_h / src_h), ncol = 2)
nepal_sf$geometry <- nepal_sf$geometry * scale_matrix

# Shift to Reference position
nepal_sf$geometry <- nepal_sf$geometry + c(target_bbox$xmin, target_bbox$ymin)

# 4. Assign CRS (Web Mercator)
st_crs(nepal_sf) <- 3857
```

```r
# Create interactive map
mapview::mapview(nepal_sf)
```

```r
# Save the georeferenced SVG map before the ward dissolve
# We use 'delete_dsn = TRUE' to overwrite any old versions
st_write(
    nepal_sf,
    rough_map_path,
    driver = "GeoJSON",
    delete_dsn = TRUE,
    quiet = TRUE
)

message("Rough SVG map saved to: ", rough_map_path)
```

6 Reconstruction via Wards

The transformed SVG is roughly accurate but has "jagged" borders and low-quality topology. To fix this, we use the **Wards** method:

1. Convert high-quality Ward polygons to centroids.
2. Check which rough SVG shape each Ward centroid falls into.
3. Dissolve (merge) the Ward polygons based on that grouping.

This results in a map that perfectly matches official administrative boundaries.

```
message("Reconstructing boundaries using Wards...")

# 1. Load Wards Data
if (!file.exists(wards_geojson_path)) stop("Wards GeoJSON not found.")
wards <- st_read(wards_geojson_path, quiet = TRUE) |>
  st_make_valid()

# 2. Project to Metric (UTM Zone 45N) for accurate cleanup
# Nepal falls mostly in UTM 45N (EPSG:32645)
wards_utm <- st_transform(wards, 32645)
nepal_utm <- st_transform(nepal_sf, 32645) |> st_make_valid()

# 3. Assign Wards to Constituencies
message("Matching wards to constituencies...")
sf_use_s2(FALSE) # Use GEOS for the join
ward_centroids <- st_centroid(wards_utm)
nearest_indices <- st_nearest_feature(ward_centroids, nepal_utm)
wards_utm$constituency_id <- nepal_utm$constituency_id[nearest_indices]

# 4. Dissolve and Clean Slivers
message("Dissolving geometries and removing slivers...")

# SNAP TO GRID: Rounds coordinates to nearest 1mm to prevent tiny float errors
# wards_utm <- st_set_precision(wards_utm, 1000)

final_map_utm <- wards_utm |>
  group_by(constituency_id) |>
  summarise(geometry = st_union(geom)) |>
  st_make_valid() |>
  # SLIVER REMOVAL: 1 meter expansion and contraction
  st_buffer(dist = 1) |>
  st_buffer(dist = -1) |>
  st_make_valid()
```

```r
# 5. Transform back to WGS84 for GeoJSON export
final_map <- st_transform(final_map_utm, 4326)
```

```r
# Create interactive map
mapview::mapview(final_map)
```

```r
# 6. Save Result
st_write(
    final_map,
    output_geojson_path,
    driver = "GeoJSON",
    layer_options = "COORDINATE_PRECISION=6",
    delete_dsn = TRUE,
    quiet = TRUE
)

message("Reconstruction Complete! Saved to: ", output_geojson_path)
```

Download the output files:

nepal_electoral_map.svg nepal_svg_rough.geojson nepal_electoral_map.geojson